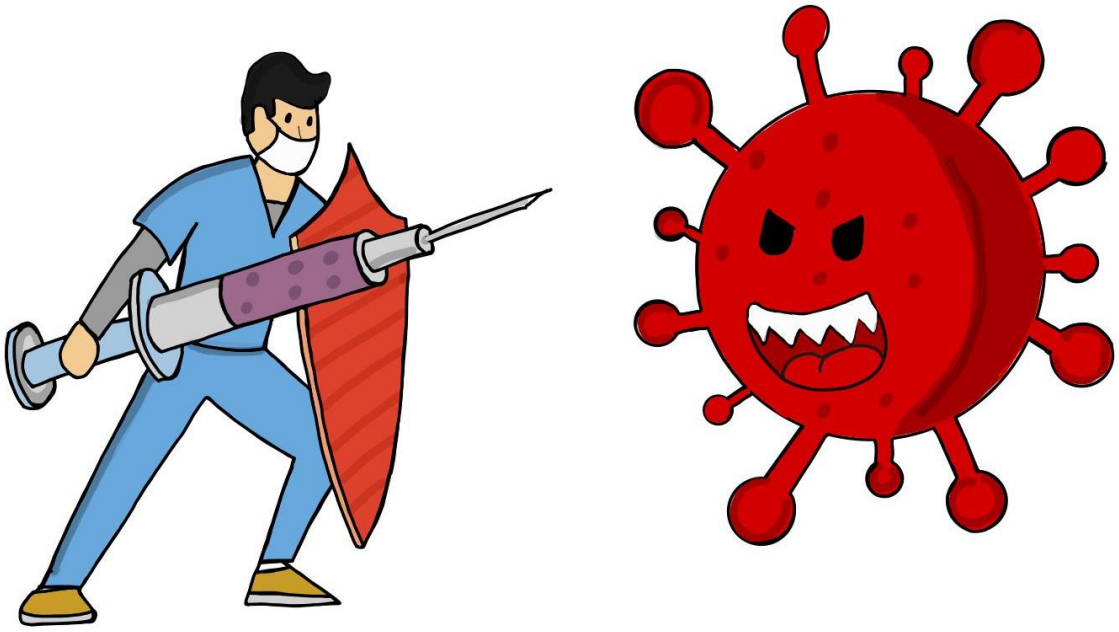
# PAEDIATRICS
# INFECTIOUS DISEASES



د. عبدالرحمن الجعكي

2025_2024

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- **Viral diseases:**
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  3. Mumps
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  6. Infectious mononucleosis
  7. Chicken pox
  8. Hand–foot & Mouth disease
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  11. **HIV infection**

- **Bacterial diseases:**
  1. Scarlet fever
  2. Whooping cough
  3. **Typhoid** fever
  4. Rickettsia –Lyme disease
  5. **Streptococcal & staphylococcus infections**

- **Protozoal Disease:**
  1. Enterobiasis
  2. Amoebiasis
  3. Giardiasis
  4. Malaria
  5. **Leishmaniasis**

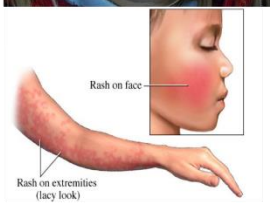
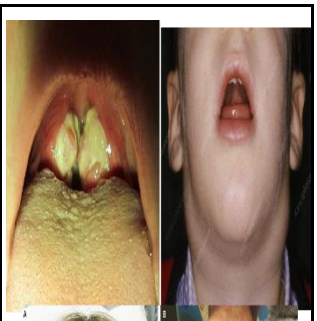
- **Tuberculosis in pediatric.**

- **PUO.**

| | Measles | Rubella | Mumps |
|---|---|---|---|
| **Etiology** | measles virus<br>RNA paramyxovirus | rubella virus<br>RNA paramyxovirus | mumps virus<br>RNA paramyxovirus |
| **Route of transmission** | droplets | droplets<br>Trans placental | droplets |
| **Incubation period** | 10–14 days (1–2wks) | 14–21 days (2–3wks) | 14–21 days (2–3wks) |
| **Infectivity period** | 5 D (rash) 4 D | 2 D (rash) 7 D | 6 D (swelling) 7 D |
| **Clinical pictures** | prodrome stage<br>C: high grade fever<br>C: cough<br>C: coryza<br>C: conjunctivitis<br>K: koplik's spots (Pathognomic)<br>Exanthem (eruptive)stage (Rash)<br>Maculopapular<br>1st behind the ears,face &head –>within 24hrs all over the body .(peelings)<br>Convalescence stage: resolution<br>  | prodrome stage<br>as measles but<br>Milder<br>• Low grade fever<br>• occipital & postauricular Lymphadenopathy.<br>• +– arthritis<br>Exanthem (eruptive) stage (Rash)<br>as measles but<br>• Less prominent<br>• Last 3 days<br>• Without peelings<br>Convalescence stage: resolution<br>  | prodrome stage<br>Mild or absent<br>Fever<br>Headache<br>Malaise<br>Parotid swelling<br>• Start uni then bilateral<br>• Better seen than felt<br>• Push ears upwards & outwards<br>• Painful & tender<br>• increases with chewing & sour<br>• Peak 3 days – subsided within 5 days<br>  |

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| **Complications** | Respiratory:<br>Otitis media**<br>Pneumonia<br>Laryngitis<br>Bronchiectasis<br><br>GIT :<br>Diarrhea + FTT<br>Appendicitis<br>Hepatitis (adult)<br><br>CNS:<br>Encephalitis<br>GBS<br>Aseptic meningitis<br>SSPE: why???<br>(1:100.000,<br>rare,late ,MRI,<br>Supportive)<br><br>Others:<br>Myocarditis<br>Corneal ulcer . | Uncommon & mild<br><br>1. Encephalitis<br>2. Neuritis<br>3. Arthritis<br>4. Myocarditis<br>5. Low plt.<br><br>Congenital rubella syndrome CRS<br><br>IUGRV<br>Microcephaly<br>MR ,D delay<br>CHD (PDA ,PS)<br>Thrombocytopenia<br>SNHL<br>Cataract,<br>glaucoma<br>Blueberry muffin rash | 1. Meningo–Encephalitis<br>2. Thyroiditis<br>3. Myocarditis<br>4. Hepatitis<br>5. Pancreatitis<br>6. Orchitis in boys (Age ??) & oophoritis in girls .<br>7. Arthritis<br>8. Transient SNHL |
| **Treatment** | Supportive TTT<br>• Good hydration<br>• Antipyretic<br>• Vit A for 2 days(decrease M&M in developing )<br>• Isolation | Supportive TTT<br>No specific treatment | Supportive TTT<br>No specific treatment |
| **Prevention** | MMR vaccine at 12–18 months of age | MMR vaccine at 12–18 months of age | MMR vaccine at 12–18 months of age |
| **Other Name** | Rubeola<br>First disease | German measles (3rd day disease) –3rd disease | |

| | Roseola infantum | Erythema infectiosum | Infectious mononucleosis |
|---|---|---|---|
| Other names | 6th disease /exanthem subitum. | 5th disease /slapped cheek syndrome | glandular fever |
| Etiology | HHV 6,7 | Parvovirus B19 | EBV |
| Incubation period | 7–15 days (1–2 wks) | 7–15 days (1–2 wks) | 30–60 days (1–2 M) |
| Route of transmission | droplets | droplets<br>Transplacental | droplets, oral contact ,blood rare! |
| Infectivity period | variable | variable | throughout the course of disease. |
| Clinical pictures | prodrome: 3 F<br>Infant well (Free)–<br>Fever (39–40 C)–<br>Febrile convulsions<br>Then<br>_____<br>Eruptive stage (exanthem)<br>On 4–5th day (maculopapular (rose)rash ,start on trunk, rapidly fade in 2 days ) | prodrome:<br>No prodrome or low grade fever, malaise, headache, and myalgia<br>Then<br>_____<br>Eruptive stage (exanthem)<br>• Slapped cheek<br>• maculopapular rash mainly in extremities & trunk fade with central clearing (lacy or reticulated)<br>• Circumoral pallor<br>• No peeling | • Fever, malaise<br>• Tonsillopharyngitis =mimic strep. (petechiae on the soft palate*)<br>• LN (cervical or generalized)<br>• Maculopapular rash (ampicillin?)<br>• Hepatomegaly 10%<br>• Splenomegaly 50% |

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| |  |    |   |
| Diagnosis | mainly clinically | mainly clinically | 1-CBC: high WBC mainly lymphocytes<br>2-Blood film: atypical lymphocytes<br>3-Monospot test (+ heterophileAb) often negative in young children<br>4-Viral capsid antigen antibodies (VCA) IgG and IgM, and EB nuclear antigen (EBNA) antibodies. |
| Treatment | Supportive TTT<br>• Good hydration<br>• Antipyretic | Supportive TTT<br>• Good hydration<br>• Antipyretic | Supportive TTT<br>• Good hydration<br>• Antipyretic<br>• Steroid if (severe upper airways obstruction)<br>• Ampicillin or amoxicillin should be avoided.?? |

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| Complication | usually is self limiting<br>But may cause :<br>aseptic meningitis, encephalitis, hepatitis, or a mononucleosis– like syndrome. | usually is self limiting & excellent prognosis<br>in children (arthritis & 8 arthralgia common in adults )<br>———<br>Other diseases caused by parvovirus are :<br>1–Transient aplastic crisis – in children chronic haemolytic anaemias<br>2–Hydrops fetalis & death due to severe anemia, although the majority of infected fetuses will recover. | Respiratory<br>Upper airways obstruction<br><br>CNS<br>Aseptic meningitis<br>Encephalitis<br>GBS<br><br>GIT<br>Hepatitis<br>Splenic rupture<br><br>others<br>Myocarditis<br>Autoimmune hemolytic anemia |

| | Chicken pox | Hand foot mouth disease | HSV Infection |
|---|---|---|---|
| Etiology | varicella zoster virus<br>VZV | Coxsackievirus (enterovirus) | HSV 1:Lips,skinn<br>HSV 2:Genetalia |
| Incubation period | 14–21 days (2–3wks) | 3–6 days | |
| Route of transmission | Droplet, direct contact ,transplacental | droplet ,feco oral | skin contact,droplets<br>Birth canal ,sexual |
| Clinical pictures | Prodrome: Fever, headache, anorexia ,malaise<br>Then<br>Rash 3P<br>Papulovesicular (in crops for 3–5 days, Pleomorphic (50–500)<br>Pruritic (itching)<br>Lesions start on head and trunk, progress to peripherals. (But maybe just a few lesions.)<br><br>Heals without scar (scar formation is secondary to bacterial infection.)<br><br>  | fever (low grade – summer)<br><br>Vesicular rash +ulcer in the hands ,feet & mout<br><br>  <br>  | asymptomatic (common)<br>Gingivostomatitis<br>Common in preschool age(10m–3yr)<br>Vesicular lesions in the<br>lips ,mouth ,tongue,hard palate<br>It's painful may lead to dehydration<br>High fever – miserable<br>May persist up to 2wks<br><br>Eczema herpiticum<br><br>  |

<table>
<tr><td></td><td></td><td></td><td>  <br>Eczema whitlows<br>Keratoconjunctivitis<br>Neonatal HSV<br>Meningoencephalitis</td></tr>
<tr><td>Diagnosis</td><td>clinically<br>PCR of vesicular fluids</td><td>clinically</td><td>clinically<br>Zank smear</td></tr>
<tr><td>Complications</td><td>2ry bacterial skin (new fever –high fever ?!) infections by staph or strep (necrotizing fasciitis–TSS)<br>CNS: encephalitis(early – good prognosis) meningitis–cerebellitis – ataxia<br>Respiratory: pneumonia<br>CVS: myocarditis, pericarditis<br>GIT: hepatitis, ulcerative gastritis, Reye syndrome<br>Renal: GN<br>Purpura fulminans Ab vasculitis in the skin and subcutaneous tissues<br>Immunocompromised<br>Haemorrhagic lesions<br>Pneumonitis<br>Progressive & disseminated infection,DIC</td><td>excellent prognosis<br>(Dehydration due to decreased oral intake )<br>Other diseases caused by enterovirus<br>Herpangina<br>meningoencephalitis<br>pleurodynia<br>myocarditis<br>enterovirus N sepsis syndrome</td><td>Gingivostomatitis<br>Good prognosis<br>Dehydration<br><br>HSV encephalitis<br>MR high<br>Neurological sequelae<br><br>HSV disseminated N sepsis (high MR)</td></tr>
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| Treatment | only supportive<br>Antipyretic (not aspirin)<br>Antipruritic<br>Acyclovir only in immune deficiency | supportive<br>Antipyretic<br>Hydration | supportive<br>Antipyretic<br>Hydration<br>Acyclovir<br>(indication ……) |
| prevention | VZ Vaccine: live attenuated<br>VZ Ig (indication …) | No vaccine | No vaccine |

## Shingles (herpes zoster)

Shingles is uncommon in children. It is caused by reactivation of latent VZV, causing a vesicular eruption in the dermatomal distribution of sensory nerves.
It occurs most commonly in the thoracic region, although any dermatome can be affected Children, unlike adults, rarely suffer neuralgic pain with shingles. Shingles in childhood is more common in those who had primary varicella zoster infection in the 1st year of life. Recurrent or multidermatomal shingles is strongly associated with underlying immunocompromise, e.g. HIV infection. In the immunocompromised individuals, reactivated infection can also disseminate to cause severe disease.

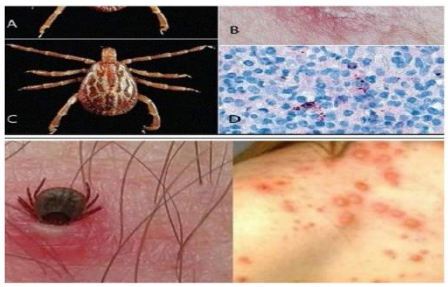


<table>
<tr><th></th><th>Hepatitis A</th><th>Acquired immunodeficiency syndrome AIDS</th></tr>
<tr><td>Etiology</td><td>hepatitis A virus (RNA, Picornavirus)</td><td>human immunodeficiency virus HIV (type 1 –type 2),RNA, retro–viridae</td></tr>
<tr><td>Incubation period</td><td>15–30 days</td><td>newborn of infected mother ≃6 months of age ,other routs may be years</td></tr>
<tr><td>Route of transmission</td><td>Feco Oral route (from person to person contact)</td><td>1–vertical **most common : (transplacental–during delivery–breast feeding)<br>2–horizontal :blood –hemodialysis – sexual contact –IV drug use</td></tr>
<tr><td>Clinical picture</td><td>asymptomatic or mild nonspecific illness is common
<table>
<tr><td>preicteric</td><td>icteric</td><td>convalescence</td></tr>
<tr><td>vomiting<br>Nausea<br>Malaise<br>Fever +–</td><td>jaundice<br>Hepatomegaly<br>Abd pain</td><td>resolution of symptoms(wks)</td></tr>
</table>
 </td><td>Asymptomatic<br>Nonspecific pictures :<br>Wt loss ,anorexia. ,fever(PUO) diarrhea,FTT,recurrent severe oral thrush ,recurrent infections(bact or viral )<br>TB,pneumocystis carinii(jiroveci),<br>Others :LD –HSM –skin rash<br>Developmental delay,HIV encephalopathy<br>recurrent parotid swelling<br><br>Clinical categories :<br>N: asymptomatic<br>A: mildly symptomatic (may only have lymphadenopathy or parotid enlargement)<br>B: moderetly symptomatic(recurrent bacterial infections, candidiasis, chronic diarrhoea, and lymphocytic interstitial pneumonitis )<br>C: severely symptomatic (opportunistic infections, e.g. Pneumocystis jirovecii (carinii) pneumonia (PCP), severe growth faltering, encephalopathy & malignancy)</td></tr>
<tr><td>Diagnosis</td><td>1–CBC –CRP .<br>2–LFT :<br>a–ALT –AST :high<br>b–ALP :normal<br>c–TB(high direct )<br>d–PT (important why??)<br><br>3–Serology :HAV specific</td><td>1–Serology :Antibodies against HIV not diagnostic early (<18 months: false +ve result ,maternal Ab!!)<br>2–DNA PCR :best diagnostic test (Two negative HIV DNA PCRs within the first 3 months of life indicate the infant is not infected)<br>3– Viral culture</td></tr>
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| | IgM(high) indicate recent infection<br>IgG (high) indicate previous infection | 4–viral load assay & CD4 count used for ? |
| **Treatment** | mainly supportive (no treatment)<br>good hydration –adequate nutrition<br>Admission if : severe vomiting dehydration –prolonged PT.<br><br>Prognosis is good (the majority of children have a mild illness and recover both clinically and biochemically within 2 weeks to 4 weeks).<br><br>Fulminant hepatic failure can occur, but it is rare (<1% and occurs more often in adults than in children | 1–Antiretroviral therapy ART :<br>(zidovudine – Lamivudine – Protease inhibition)<br>depends on clinical status, HIV viral load, and CD4 count, except in infants who should all start ART shortly after diagnosis.<br>2–. Prophylaxis against PCP with cotrimoxazole .<br>3–aggressive treatment of infection<br>4–Nutritional support<br>5–Multidisciplinary team<br>6–Vaccination schedule !?!<br>7–Regular follow–up, with particular attention paid to weight, developmental progress, and clinical signs and symptoms of disease. |
| **Prevention** | 1–Hygienic measures (hand washing)–food sanitation<br>2–HAV vaccine to close contact<br>3–isolation (duration ?) | Decrease risk of vertical transmission to (<1%) by the following:<br>1–Maternal ART.<br>2–Elective C/S, avoid prolonged rupture of the membranes and unnecessary instrumentation .<br>3–Zidovudine for infant (6wks )<br>4–Contraindication of breastfeeding |

| | Scarlet fever | Pertussis (whooping cough) |
|---|---|---|
| Etiology | Group A beta hemolytic streptococci (+ve ): pyrogenic exotoxin | Bordetella pertussis (–ve coccobacilli),Endemic, with epidemics /3–4 years. |
| Route of transmission | Droplets | Droplets<br>Highly contagious |
| Incubation period | 2–7 days | 7–10 days (5–21days) |
| Clinical picture | 1–Fever –sore throat,headache,malaise<br>2–Tongue : 1st day (white strawberry<br>3rd day (red strawberry)<br><br>3–Rash (12–48 hrs of onset )<br>First in the neck –>trunk & extremities (diffuse maculopapular)<br>:Sandpaper palpation .<br>Prominent in flexor skin creases ,cubital fossa :pastia lines .<br>Last for ≃4–5 days with desquamation<br>4–Face:red +circumoral pallor<br><br>  <br>  <br>  | Catarrhal stage (1–2wks)<br>Low grade fever –coryzal symptoms<br><br>Paroxysmal stage (2– 4 to 6 wks)<br>Attacks of spasmodic cough in expiration followed by inspiratory whoop & posttussive vomiting<br>During a paroxysm, the child goes red or blue in the face, and mucus flows from the nose and mouth.<br>(In young infant & preterm .....may be absent & .....is common feature)<br><br>Convalescence stage (1–2wks)<br>Cough episodes decreased in severity number gradually<br>May persist up to 3 months (100 day cough ) |
| Diagnosis | • Clinically<br>• Serology (ASO titre – antiDNAase Ab )<br>• Throat swab C&S . | • CBC :high WBC (absolute lymphocytosis )<br>• Nasopharyngeal swab for PCR ,ELISA .<br>• Chest x ray |

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| Complication | • PSGN<br>• Reactive arthritis<br>• Rheumatic fever<br>• Cervical LD<br>• Otitis media | Respiratory:pneumonia–OM–Apnea–Atelectasis–Bronchiectasis–Pneumothorax<br>CNS: Convulsion–IC hemorrhage Encephalopathy Retinal or Subconjunctival Hge<br>GIT:Hernia–Rectal prolapse–Malnutrition |
| Treatment | • Antipyretic<br>• Penicillin V for 10 days | If severe or apnea ,very young or PT: Admission<br>• O2 therapy if low O2 Saturation<br>• NGT feeding or IVF<br>• Macrolides (DOC ) |
| prevention | • No vaccine | • Isolation is recommended.<br>• Macrolides to all contacts.<br>• DPT vaccine (Unimmunized infant contacts should be vaccinated). |

| | Rickettsia (Rocky Mountain spotted fever) | Typhoid fever (enteric fever) |
|---|---|---|
| **Etiology** | R.rickettsii (–ve obligatory intracellular) | Salmonella typhi or paratyphi (–ve ) |
| **Route of transmission** | Vector (Tick)<br>Reservoir (dogs) | Feco–oral :street food or contaminated water |
| **Incubation period** | 2–14 days (≃10days) | 3–30 days (1–2wks) |
| **Clinical picture** | 1–Non specific symptoms:fever high grade –headache–malaise<br>2–Rash: ≃90%<br>Maculopapular all over the body including palms & soles<br>Petechial or purpuric rash !?<br><br>3–Other manifestation splenomegaly–myocarditis–pneumonitis –renal impairment & shock.<br>  <br>  | 1–Fever : high spiking &(relative bradycardia)<br>2–headache,myalgia,cough<br>3–Abdominal pain,anorexia<br>3–Diarrhea +– bl then constipation (in 2nd week)<br>4–Rash (ROSE SPOTS on trunk )<br>5–Hepatosplenomegaly<br><br>If no complications manifestations gradually resolve within 2–4 wks<br>  |
| **Diagnosis** | • Clinically<br>• CBC :anemia – thrombocytopenia<br>• Electrolyte:hyponatremia<br>• LFT : high ALT ,AST<br>• Serology test & PCR | • CBC : high or low WBC ,N or low plat<br>• Blood culture:1st week (+ 40–80% early)<br>• Stool&urine |

| | | |
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| | | culture:2nd week<br>• BM culture:!<br>• Widely test :?!<br>• PCR detection of typhi antigen |
| **Complication** | 1-Shock ,DIC ,low BP<br>2. Acute Renal failure<br>3. Hepatitis<br>4. Meningoencephalitis<br>5. ARDS ,multiorgan failure | 1-Intestinal Hge & perforation <1% .<br>2-Osteomyelitis<br>3-GN<br>4-myocarditis<br>5-Hepatitis<br>6-Neurological complication:as delirium,ataxia & GBS uncommon |
| **Treatment** | • Admission in severe cases<br>• IVF +antipyretic<br>• Antibiotics<br>○ <8yr doxycycline<br>○ >8yr tetracycline<br>○ Macrolides Azithromycin | • most of cases treated at home<br>• Admission:in severe cases<br>• IVF +antipyretic<br>• Antibiotics IV for 2 weeks<br>○ 3rd generation cephalosporin -azithromycin are effective |
| **prevention** | no vaccine | 2 vaccines are available (live attenuated & killed type) |

## Lyme disease

**Cause:** Spirochaete (Borrelia burgdorferi)
**Route of transmission:** Tick bite (host including sheep, foxes and many small mammals)
**I.P:** 4–30 days (as typhoid fever **)
**Season:** more in summer and rural area

**C/P:**

1. Erythema migrans – Erythematous macule painless at the site of tick bite enlarges and cause this classical sign



2. Fever – headache , myalgia – arthralgia
3. Lymphadenopathy (painfull)
4. May lead to cranial nerve palsy – meningitis – carditis
5. The late stage: weeks to months
    - CNS – meningoencephalitis – facial nerve palsy – peripheral neuropathy
    - CVS – Myocarditis –HB
    - Joint – 50% (arthralgia – large joint, recurrent arthritis, erosive arthritis 10%)

**Diagnosis:** Clinically mainly – Serology (+ve late in diseases > 4wks)
**Treatment:** Doxycycline DOC ( >8year) & Amoxicillin DOC in young child

- IV 3rd generation cephalosporin if severe carditis or neurological disease

## Staphylococcus & streptococcal disease

Diseases caused by Staph. areus
Sepsis–Suppurative arthritis & osteomyelitis–Impetigo, folliculitis, fruncles, carbuncle, –Parotitis, Membranous tracheitis (complicating viral croup)–Pneumonia
Diseases caused by toxins

- Staphylococcal Food poisoning

**Cause:** Staph. enterotoxin in contaminated food – **Source:** food handlers
**Clinical:** start after 2–7 hrs, vomiting, water diarrhea, last for 12–24 hours,No fever
**Diagnosis:** Culture & testing for enterotoxin of suspected food
**Prevention:** Exclude infected food handler, eat foods immediately or refrigerate it

| Toxic shock syndrome TSS | Staphylococcal scalded skin syndrome (Ritter disease) |
|---|---|
| **Cause:** Toxic Shock Syndrome Toxin–1 (TSST–1)<br>( S. aureus & group A streptococci)<br><br>**Clinical picture :**<br>Major criteria (all required):<br>– Acute fever > 39 C<br>– Hypotension (may progress to Shock & DIC)<br>– Skin rash: diffuse erythroderma (sunburn like) with desquamation (palms & soles)<br>Minor criteria (any 3 or more):<br>– Inflamed mucous membrane (vaginal, oropharyngeal, conjunctival, strawberry tongue) – –GIT abnormalities (Vomiting or diarrhea)<br>– Liver abnormalities<br>– Renal abnormalities (BUN or Creatinine high)<br>– Muscle abnormalities (Myalgia or CPK > twice upper normal limit)<br>– CNS abnormalities (Altered consciousness without focal neurological signs) – Thrombocytopenia<br>Exclusion criteria<br>– Absence of another explanation – Negative blood culture<br><br>**Complications:**<br>– ARDS, Myocardial infarction, Renal failure<br>– Hair & nail loss after 1–2 mo<br>– Recurrence if inadequate treatment<br><br>**Treatment**<br>– Supportive: fluid, intropes<br>– Specific:3rd generation cephalosporin + clindamycin<br>– Surgical drainage of infected sites<br>– Severe cases: IVIG | **Cause:** Staph. exfoliative toxins A & B<br><br>**Clinical:**<br>If neonate → Ritter diseases (generalized exfoliation)<br>If infant → white/brown flaky desquamation of the entire skin especially face & neck<br>Cleavage of superficial layer of epidermis → healing without scarring<br>Nikolsky sign + :separate on gentle pressure.<br><br>  <br><br>**Complications:**<br>dehydration, superinfection<br>If child → localized bullous impetigo + tender scarlet like eruption<br><br>**Treatment:**<br>intravenous antistaphylococcal antibiotic (e.g. flucloxacillin), analgesia, and monitoring of hydration and fluid balance. |

## Impetigo:

**Definition :** is a localized, highly contagious, staphylococcal or streptococcal skin infection

**Age :** infants and young children , is more common in children with pre-existing skin disease, e.g. atopic eczema.

**Sites :** usually on the face, neck, and hands.

**Clinical picture:** begin as erythematous macules that may become vesicular/pustular or even bullous Rupture of the vesicles with exudation of fluid leads to the characteristic confluent honey-coloured crusted lesions. Infection is readily spread to adjacent areas and other parts of the body by autoinoculation of the infected exudate.



**Treatment:** Topical antibiotics (e.g. mupirocin) are sometimes effective for mild cases. Narrow-spectrum systemic antibiotics (e.g. flucloxacillin) are generally needed for more severe infections.
Affected children should not go to nursery or school until the lesions are dry.

## Periorbital cellulitis:

**Definition:** In periorbital (or preseptal) cellulitis there is fever with erythema, tenderness, and oedema of the eyelid or other skin adjacent to the eye and anterior to the orbital septum.

**Site :** It's almost always unilateral.

**Treatment:** should be treated promptly with iv antibiotics such as ceftriaxone to prevent posterior spread of the infection to the orbital tissues causing orbital cellulitis.



In orbital cellulitis:there's proptosis, painful or limited ocular movement with or without reduced visual acuity)

**Complications:** by abscess formation, meningitis, or cavernous sinus thrombosis. Where orbital cellulitis is suspected, a CT or MRI scan should be performed to assess the poste- rior spread of infection and managed in conjunction with an ophthalmologist.

## Causes of fever and a rash: mcq

| Maculopapular rash | Vesicular, bullous, pustular | Petechial, purpuric rash |
|---|---|---|
| **Viral**<br>1–(HHV–6) or HHV–7 (roseola infantum) – <2 years old<br>2–Enterovirus rash<br>3–Parvovirus (slapped cheek) usually school age<br>4–Measles &rubella – uncommon if immunized<br><br>**Bacterial**<br>1–Scarlet fever (group A streptococcus)<br>2–Toxic shock syndrome<br>3–Rheumatic fever – erythema marginatum<br>4–Salmonella typhi (typhoid fever) – classically rose spots<br>5–Lyme disease – erythema migrans<br><br>**Others**<br>1–Kawasaki disease<br>2–Systemic onset juvenile `idiopathic arthritis | **Viral**<br>1–Varicella zoster virus – chickenpox, shingles<br>2–Herpes simplex virus<br>3–Coxsackie virus – hand, foot and mouth<br><br>**Bacterial**<br>1–Impetigo – characteristic crusting<br>2–Boils<br>3–Staphylococcal bullous impetigo<br>4–Staphylococcal scalded skin<br><br>**Others**<br>1–Erythema multiforme/ Stevens–Johnson syndrome/ toxic epidermal necrolysis | **Bacterial**<br>1–Meningococcal, other bacterial sepsis<br>2–Infective endocarditis<br><br>**Viral**<br>1–Enteroviruses, adenoviruses, and other viral infections<br><br>**Other**<br>1–Henoch–Schonlein purpura<br>2–Thrombocytopenia as idiopathic thrombocytopenic purpura ITP<br>3–Vasculitis<br>4– Malaria |

# Parasitic infestation

## Enterobiasis

**Cause:** Enterobius vermicularis (Pinworm) "Nematode" ,:A small (1 cm in length), white, threadlike nematode (roundworm), that typically inhabits cecum, appendix, and adjacent ileum and ascending colon.

**Transmission:** From person to person via Feco – oral route،auto infection may occur in individuals who habitually put their fingers in their mouth.

Age: School and preschool age groups

**Sex:** Male = Female, all socioeconomic classes are affected

**C/P:**

- Asymptomatic
- Symptomatic (anal itching increase night, sleep disturbances , restlessness and enuresis)

**Diagnosis:**

- Transparent tape (adhesive cellophane tape) is pressed against the perineum at night and examined at morning – adult worm or its egg seen.

**Treatment:** Infected individuals and their family members should be treated.

- Albendazole (400 mg PO with a repeat dose 2 wk later for all age groups).or
- Mebendazole (100 mg PO with a repeat dose 2 wk later)

**Prevention:**

- Good hand hygiene is the most effective method of prevention.
- Household contacts can be treated at the same time as the infected individual.

## Amoebiasis

**Cause:** E. histolytica

**Transmission:** Feco – oral route : Food or drink contaminated with Entamoeba cysts which are resistant to harsh environmental conditions, including chlorine commonly used in water purification, but can be killed by heating to 55°C). After ingestion, cysts are resistant to gastric acidity and digestive enzymes and form trophozoites which colonize large intestine lumen and may invade the mucosal lining.

**C/P:**

- Asymptomatic carriers
- Amebic dysentery 5–10%
  - Common in 1–5 yr of age and occur within 2 wk of infection (or may be delayed).
  - Gradual onset of colicky abdominal pain and frequent bowel movements (diarrhea 6–8/day), frequently associated with tenesmus , stool is heme–positive
  - Generalized constitutional symptoms are absent (fever occurs in only ⅓ of patients).
- Hepatic abscess ≈1%

- Uncommon in children/ may occur months to years after exposure.
- May appear in patients with no clear history of intestinal disease.
- Fever is frequently associated with abdominal pain, abdominal distension, and enlargement and tenderness of the liver + right-sided atelectasis or effusion.

**Diagnosis:**

- Stool microscopic examination cyst (3 samples :sensitivity 80-90%)
- Stool for E histolytica antigen or PCR .
- In case of liver abscess:high WBC ,ESR,hepatic enzymes ,Ultrasonography, CT, or MRI can localize and determine the size of the abscess cavity.(common is single, in the right lobe ).

**Treatment:**

- Symptomatic cases –Metronidazole DOC or tinidazol
- Asymptomatic cases – Paromomycin – iodoquinol to eradicate infection

**Prevention**

- Proper sanitation and avoiding fecal-oral transmission.
- Regular examination of food handlers and thorough investigation of diarrheal episodes may help identify the source of infection.

## Leishmaniasis

Multiple species of leishmania : cutaneous disease is generally mild or mucosal and visceral leishmaniasis is associated with significant morbidity and mortality.

Mode of transmission – insect by ♀ sand flay

**C/P:**

1. Cutaneous leishmaniasis : L. major, L. Tropica, L. Mexicana
   - C/P: On exposed area – papule – nodule – ulcer painless – heal with scare .
2. Visceral Leishmaniasis (Kalaazar): Age < 5 years – L. chagasi, L. Infantum, L. donovani. :Prominent reticuloendothelial cell hyperplasia in liver, spleen, BM, and LN.
   - asymptomatic or have (mild malaise, intermittent diarrhea, poor activity tolerance, and intermittent fever/ most will have a mildly enlarged liver).
   - Most cases resolve without therapy/ 25% evolve to active kala-azar within 2-8 mo.
   - Classic clinical features include:[High fever–marked splenomegaly–Hepatomegaly–Severe cachexia]
   - At terminal stages of kala-azar:
     - Massive hepatosplenomegaly & Profound pancytopenia:Anemia may be severe enough to precipitate heart failure.
     - Bleeding episodes, especially epistaxis, are frequent.
     - Secondary bacterial infections, which frequently are a cause of death.

- Gross wasting, jaundice, edema, and ascites.
- Death occurs in > 90% of patients without specific anti-leishmanial treatment.

Rx:

- Sodium stibogluconate ([Pentostam)
- Amphotricin B

## Malaria

Malaria should be considered in any febrile child who has returned from a malaria-endemic area within the previous year (Africa, Asia, and South America)& most malarial deaths occur among infants and young children.

- Infecting agent – Sporozoites
- Vector: Infected ♀anopheles mosquitoes
- Host and Reservoir – Human (RBC – Liver)
- Mode of transmission (Female Anopheles mosquitoes (Vector)–Blood transfusion and contaminated needles–Transplacentally from a pregnant woman to her fetus)

Types of malaria:

- Plasmodium falciparum (Acute – Fatal)
- Plasmodium vivax (commonest)
- Plasmodium ovale
- Plasmodium malariae (mild but chronic)

Incubation period: (9 –40 days)

C/P: Malaria is characterized by paroxysms of fever, chills, sweats, anemia, and splenomegaly.

- Fever (High grade) intermittent
- Anemia (Hemolysis)
- Splenomegaly
- Jaundice (mild – sever)
- headache, vomiting, diarrhea,fatigue, anorexia, myalgia, and pain in the chest, abdomen, and joints.

P. Falciparum:parasitemia, High MR

- Severe anemia–jaundice –splenomegaly –headache–Fever .
- Cerebral malaria: Drowsiness – Convulsion – Coma
- Hypoglycemia in cerebral malaria –metabolic acidosis.
- Microthrombosis–hemiplegia – GI Bleeding –renal failure–pulmonary edema(RD) are rare .

Diagnosis:–

- CBC (Hb low , WBC, PH )
- Blood film thick and thin film (Giemsa–stained smears) diagnostic **, A single negative blood smear does not exclude malaria & smears should be obtained several times a day for 3 days.

- Serology ELISA
- LFT.
- IG level..... ?

**$R_x$:**

1. Supportive $R_x$ (Antipyretic – Rehydration –$R_x$ convulsion – $R_x$ Hypoglycemia– $R_x$ of anemia – Hypotension
2. Drugs:Chloroquine phosphate–Quinine sulfate– Primaquine (Prevent relapse)

**Prophylaxis:**

- Chloroquine (Traveling to an endemic area), 2 weeks before travel & 8 weeks after return
- no efficient vaccine is available (under trial)

**Prevention of malaria:**

Human:

1. $R_x$ index point and carrier
2. Prevent biting by
   - Covering up the exposed skin
   - Use of insect repellents
   - Use of mosquito net over bed

Vector control:

- $R_x$ of breeding places
- $R_x$ of breeding places with larvicidal residual insecticides

Note : RBCs containing HbS (sickle erythrocytes) resist malaria growth while RBCs containing HbF (fetal hemoglobin) and ovalocytes are resistant to P. falciparum.

**Congenital malaria:** Acquired from the mother prenatally or perinatally

- May cause abortions, stillbirths, premature births, IUGR, and neonatal deaths, even in the absence of transmission from mother to child.
- Manifests between 10–30 days of age (range: 14 hr to several months of age) with:
  - Fever, restlessness, drowsiness, cyanosis, and poor feeding
  - Pallor, jaundice, and hepatosplenomegaly
  - Vomiting and diarrhea

# Tuberculosis in children

## Tuberculosis

**Etiology**: Mycobacterium tuberculosis (TB Bacilli:Aerobic,Acid–fast bacilli,Intracellular,Stained by Ziel–Neelsen stain, Culture: Lowenstein–Jensen medium)
**Route of Transmission:** By droplet routes from person to person
Incubation period: 6 weeks with insidious onset
**Risk factor:**

- Poor socio – economic condition
- Poverty & malnutrition
- Child contact with infected adult
- multidrug–resistant Mycobacterium tuberculosis strains (MDR–TB)
- Immunodeficiency (HIV)
- Travel to endemic area

| Pediatric TB vs Adult TB |
| --- |
| 1–Latent TB .....? to active TB in infants and young children compared with adults.<br>2–children are generally less ....., because their disease is typically paucibacillary. |

**Clinical pictures:**
1. Asymptomatic cases

- 50% of infants – 90% of old child
- Discovered during routine testing (TST)
- Should receive chemoprophylaxis (IGRA)

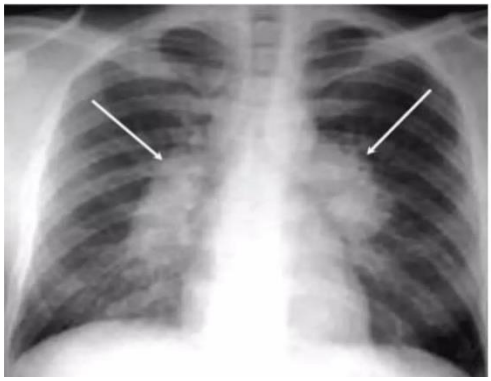
2.Primary T.B : lungs most common site ,(other less common sites :lymph node ,skin ,intestines )

- Lead to formation of 1ry complex:Ghon's Focus,TB lymphangitis, TB lymphadenitis (Hilar LN)
- Fate of the 1ry complex:1–Good immunity: Healing + Fibrosis + Calcification but.
  2– Poor immunity: Active 1ry disease + Spread:–
    - Direct: Pneumonia, cavitation, pleurisy, pleural effusion,collapse
    - Blood: isolated organ or miliary TB
    - Bronchial: Larynx, pharynx, tongue, intestine
    - LN enlargement: Pressure manifestations

| Latent TB | Active TB |
|---|---|
| • TB germs are alive but "asleep" in your body<br>• Can't spread the disease<br>• No symptoms<br>• TST & IGRA :Positive<br>• Normal chest x-ray<br>• Can progress to active TB disease (especially if infant & child < 4yrs , co-infected with HIV ,immune compromised. Mcq) | • TB germs are alive and "awake" in your body<br>• Can spread the disease<br>• Has symptoms<br>• TST & IGRA :Positive<br>• Abnormal chest x-ray<br>• Can lead to death if left untreated |

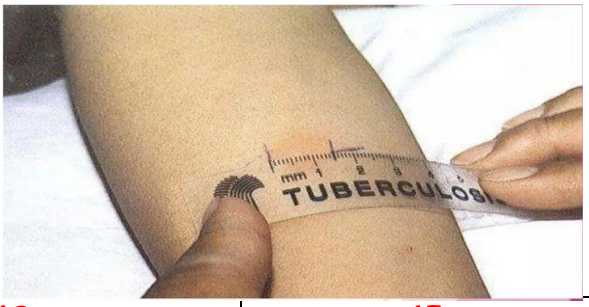
- Pulmonary T.B
  - Contact with affected family member
  - General (TB toxemia): Night fever, night sweat, anorexia, loss of weight
  - Chronic cough: is the main symptom with or without expectoration
  - Sputum may be mucoid, purulent or bloody sputum
  - Chest pain, dyspnea or wheezes may occur
  - Pneumonia, pleural Effusion
- Extrapulmonary TB
  a. TB Lymphadenitis:Cervical LN , firm, discrete, mobile, non-tender
  b. Urogenital TB:Renal TB results from hematogenous spread
  c. CNS TB: common < 5yr with high mortality & morbidity, include TB meningitis 0.3% (increased ICP, Irritability, Convulsions...), CSF: high Protein, high cells (mainly lymphocytes), low Glucose, Tuberculoma:Formation of large caseous lesion, Space-occupying manifestations
  d. Skeletal TB:Pott's disease of the spine,Site: Lower thoracic (Commonest site), Lumbar, Cervical
  e. Abdominal TB:TB enteritis:Chronic diarrhea, tenesmus, bleeding, TB peritonitis, ascitis .
  f. TB arthritis:Site: Hip, Ankle, Knee, Elbow joints, C/P: Pain, swelling, limitation of movements, X-rays: Bone rarefaction & destruction
  g. Other types:TB pericarditis, TB skin Site: Chin, lips, nose, limbs, genitalia (Lupus vulgaris)
  h. Miliary TB :Wide spread hematogenous dissemination to multiple organs with hepatosplenomegaly.

- **Investigations**



  a. CBC: Lymphocytosis , ESR: high , CRP: Positive
  b. Imaging: CXR or CT chest: (hailer LD ,collapse ,consolidation,effusion or miliary TB.
  c. Sample: Sputum, stomach wash, gastric aspirate , direct smear: Ziel–Neelsen stain for acid fast bacilli .
  d. Culture: Lowenstein–Jensen medium (requires 3–6 wks to give results)
  e. New : Gene x pert study , Whole genome sequencing of samples,PCR : provides_ genotypic data on mutations associated with resistance.
  f. Tuberculin test(Mantoux test)
     - Principle: Delayed hypersensitivity reaction (Type IV) to TB bacilli
     - Method: –Intra–dermal injection of 0.1 ml containing 5 units of purified protein derivative (PPD) in the skin of the flexor surface of the forearm
     - Area of induration (Not erythema) is measured 48–72 hrs after administration
     - Both longitudinal & transverse diameters are measured (The mean is recorded)
     - Interpretation: The reaction should be measured in millimeters of the in duration






| ≥5 mm | ≥10 mm | ≥ 15 mm |
|---|---|---|
| Children with close contact to infectious TB | Children & infants with increased risk of disseminated disease:(< 4 yrs ,DM,CKD,Lymphoma …) | Children with No risk factors |
| Children with clinical evidence of possible TB | Children increased risk of exposure to TB | |
| Children with radiological feature of TB | | |
| Children with immune suppressive (including HIV) | | |

## Causes of false results of TST:

| False +ve | False –ve |
|---|---|
| • BCG vaccination<br>• Immune reactivity to non TB mycobacteria | • Malnutrition,very young age .<br>• HIV (Immune deficiency )<br>• Reading error<br>• Improper ID injection of PPD<br>• Disseminated TB disease<br>• Recent viral infection or MMR vaccine |

g. Interferon–gamma release assays (IGRAs) are newer, blood–based tests for TB. They assess the response of T cells to in vitro stimulation, with a small number of antigens expressed by M. tuberculosis but not by BCG. Positive results therefore indicate TB infection rather than BCG vaccination. However, a negative IGRA result does not reliably rule out TB infection.

- Treatment:
  - Early detection and immediate treatment is important
  - In case of pulmonary T.B "6m": (2 months: Isoniazid, rifampicin, pyrazinamide, ethambutole –> Then 4 months: Isoniazid and rifampicin)
  - Miliary TB, bone,joints & TB meningitis "9–12m": (3 months: Isoniazid rifampicin and pyrazinamide—> then 9 months: Isoniazid and rifampicin
  - In tuberculous meningitis, dexamethasone is given initially, to decrease the risk of long–term sequelae.

- Prevention of TB
- BCG Vaccine (Bacillus–Calmette–Guerin] mcq
  - Nature: Live attenuated, dose: 0.1 ml (0.05 ml in neonates)
  - Route: Intradermal in the Lt deltoid area
  - Timing: Starting at birth, Booster doses may be needed .
  - Contraindications: Immunodeficiency, immunosuppressives
  - Advantages: Prevention of pulmonary TB (≃ 50% effectiveness) & Prevention of TB meningitis & disseminated disease (*50–80% effectiveness)
  - Side effects (Reaction):Erythema, papule formation = ulceration in 3–6 wks– Scar formation within 2–6 months– Regional lymphadenitis– Disseminated BCG infections in immunocompromised individuals
- Chemoprophylaxis
  - Indications: High–risk close contacts
  - Dose: Isoniazid (INH).

## Pyrexia of unknown origin. "PUO"

**Definition:** For clinical purposes, we define FUO in children as fever >38.3°C at least once per day for ≥8 days ( >1wk in infants &young children &>2–3wks in adolescent) with no apparent diagnosis after initial outpatient or hospital evaluation that includes a detailed history, thorough physical examination, and initial laboratory

**Causes:**

| Infection | TUMORS | CTD | Miscellaneous |
|---|---|---|---|
| Bacterial: TB– IE – UTI – Typhoid fever–brucellosis– cat scratch disease – osteomyelitis–septic arthritis,deep abscess as liver.<br>Viral: HIV – EBV – CMV– adenovirus<br>Parasite: Malaria– toxoplasmosis | Leukemia<br>Lymphoma<br>Neuroblastoma<br>Sarcoma | juvenile idiopathic arthritis<br>Kawasaki disease<br>SLE<br>Rh fever | • Diabetes insipidus (central or nephrogenic)<br>• Drug fever<br>•Factitious(Fabricated : Munchausen syndrome by proxy).<br>• Familial dysautonomia<br>•Immunodeficiency<br>• Inflammatory bowel disease<br>• Kawasaki disease<br>• Periodic fever (eg, Mediterranean fever,) |

**Management:**

- Good history & examination is the most Important point .
- Investigation – CBC – CRP–PCT –ESR– Urine C&S – TB skin test –viral serology test – X Ray – BMA …ect

$R_x$ Admission for observation & antibiotic use may obscure the diagnosis .

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3. Nelson essentials of pediatrics 9th edition.
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تنبيه

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